Network Topologies

What it is

Network topology defines the physical and logical arrangement of nodes and links in a network. Physical topology describes actual cabling and device placement; logical topology describes data flow paths regardless of physical layout. Both determine failure domains, bandwidth sharing, and routing complexity.

Why it matters

Topology directly constrains latency bounds, fault tolerance, and scaling characteristics — choices made at design time propagate to every protocol layer above. Misaligned topology causes broadcast storms, single points of failure, or $O(n^2)$ cabling costs that no protocol tuning can fix.

How it actually works

Bus — All nodes share a single coaxial or twisted-pair backbone (10BASE2/10BASE5). CSMA/CD governs access: nodes transmit when carrier sense detects idle; collisions trigger exponential backoff (binary exponential backoff, max 16 attempts per IEEE 802.3). Terminators (50 Ω) at both ends prevent signal reflection. Maximum segment length: 185 m (10BASE2) or 500 m (10BASE5). Bandwidth is half-duplex, shared — aggregate throughput drops linearly with node count.

Star — Each node connects point-to-point to a central switch/hub via dedicated link (typically UTP Cat5e+). Switches operate at Layer 2, learning MAC addresses via source inspection, building forwarding tables (CAM table). Full-duplex operation eliminates collisions; no CSMA/CD needed. Failure domain limited to single link. Uplink bandwidth between switches becomes bottleneck — typically aggregated via LACP (IEEE 802.3ad). Maximum cable run: 100 m per TIA/EIA-568.

Ring — Nodes form closed loop; each forwards frames to next neighbor. Token Ring (IEEE 802.5) uses 3-byte token circulating at 4/16 Mbps; only token holder transmits. FDDI (ANSI X3T9.5) dual-ring at 100 Mbps over fiber — primary ring active, secondary provides wrap-around on failure (50 ms typical). Deterministic latency: max wait = token rotation time (TRT). No collisions, but single break partitions network unless dual-ring.

Mesh — Every node connects to every other (full mesh: $n(n-1)/2$ links) or subset (partial mesh). Routing uses link-state (OSPF, IS-IS) or distance-vector (BGP) protocols. Full mesh provides $n-1$ disjoint paths — survives any $n-2$ failures. OSPF SPF calculation $O(E \log V)$ per area; LSDB sync floods LSAs every 30 min (refresh) or on change. BGP path-vector avoids loops via AS_PATH attribute. Partial mesh trades path diversity for link count — common in WAN/backbone.

Hybrid — Combinations: star-of-stars (core/distribution/access layers), star-ring (FDDI backbone with star drops), mesh-star (data center spine-leaf). Spine-leaf: every leaf connects to every spine (full bipartite). ECMP (Equal-Cost Multi-Path) hashes flows across spines — 5-tuple hash (src/dst IP, src/dst port, protocol) per RFC 2992. Oversubscription ratio (typically 3:1) determines leaf-to-spine uplink count.

Architecture / flow

```
Host --(100m UTP)--> Access Switch --(10/40/100G)--> Spine Switch --(ECMP)-
-> Remote Leaf --> Host
                   |                    |
                   +-- L2 domain        +-- L3 routed (OSPF/IS-IS/BGP)
                   |  (STP/RSTP)        |  (ECMP hash)
                   v                    v
              Broadcast domain      No broadcast
              limited to VLAN       across spines
```

Key terms

Collision domain — Segment where CSMA/CD contention occurs; eliminated per-port by switches. **Broadcast domain** — Set of nodes receiving Layer 2 broadcasts; bounded by VLANs (IEEE 802.1Q). **Spanning Tree Protocol (STP/RSTP/MSTP)** — Loop prevention in L2 topologies; RSTP (802.1w) converges in ~1 sec vs 30-50 sec. **ECMP** — Per-flow load balancing across equal-cost paths; hash stability requires consistent 5-tuple. **Oversubscription ratio** — Ratio of downlink to uplink bandwidth; 3:1 typical for spine-leaf. **Token rotation time (TRT)** — Max latency bound in token ring; TRT = ring latency + sum of node hold times.

Example

```
# Spine-leaf ECMP verification (Linux)
ip route show 10.0.0.0/8
# 10.0.0.0/8 nexthop via 10.1.1.1 dev eth1 weight 1
#     nexthop via 10.1.2.1 dev eth2 weight 1
#     nexthop via 10.1.3.1 dev eth3 weight 1
# Demonstrates equal-cost multipath across 3 spine uplinks
```

Common mistakes

- Assuming star topology eliminates all contention — head-of-line blocking at switch egress queues still causes latency spikes under oversubscription.
- Configuring STP priorities without root bridge placement — default bridge ID (32768 + MAC) often elects suboptimal root, creating asymmetric paths.
- Using LACP without matching hash algorithms on both ends — causes flow polarization (all traffic on one physical link).
- Deploying full mesh in large clusters — $O(n^2)$ links exceed switch port density; partial mesh with route reflectors (BGP) or areas (OSPF) required.

Lesser-known internals

- **RSTP proposal/agreement handshake** converges point-to-point links in 3 BPDU exchanges (~100 ms) but falls back to 802.1D timers on shared-media ports — half-duplex links disable fast transition.
- **ECMP hash polarization** — identical 5-tuple hashes across multiple tiers cause flow collapse; mitigate with per-tier hash seeds (RFC 7424) or adaptive hashing.
- **FDDI dual-ring wrap** — on single break, stations adjacent to break loop back; wrap completes in <50 ms but doubles traffic on remaining ring — capacity planning must account for 2x load during failure.

Learn next

- **TCP/UDP** — Transport protocols operate atop topology; congestion control reacts to loss patterns topology creates.
- **Load balancing** — ECMP and LACP are topology-aware load distribution; higher-layer LB (L4/L7) builds on these primitives.
- **Distributed Systems: CAP theorem** — Network partitions are topology failures; topology design defines partition probability and blast radius.